CDC BRFSS 2015 · RESEARCH PROTOTYPE

Early Diabetes Risk Screening

Can self-reported health answers find adults with diabetes or prediabetes, while keeping missed cases low?

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THE QUESTION

A screening tool should miss few cases without flagging everyone

253,680

adults in the CDC's 2015 BRFSS survey

21

self-reported answers each: health, habits, age, income

15.8%

report diabetes or prediabetes

The bar to clear: not random guessing, but the paper Diabetes Risk Test that clinics already use (NIDDK/ADA).

THE DATA

We count prediabetes as positive, on purpose

Survey status	Respondents	Main analysis	Prediabetes only
No diabetes	213,703	negative	negative
Prediabetes	4,631	positive	positive
Diabetes	35,346	positive	excluded

Main: 39,977 positives (15.8%). Prediabetes is the state a screening tool exists to catch.

Sensitivity check: prediabetes alone, 4,631 of 218,334 (2.1%).

EXPLORATION AND TESTS

Five answers carry the signal; one mostly measures seeing a doctor

Strongest correlations with the outcome

Answer	Spearman
General health	0.30
High blood pressure	0.27
BMI	0.24
Difficulty walking	0.22
High cholesterol	0.21

Unadjusted odds ratios

High blood pressure: 4.8× the odds

95% CI 4.67–4.90

Cholesterol check: 5.9×, the largest

Mostly contact with doctors, who make the diagnosis

All 25 planned tests significant after Benjamini–Hochberg correction: with 250,000 people, read effect sizes, not p-values · 4

EVALUATION DESIGN

Every decision used training rows; the test set stayed locked

Training: 202,944 people, five fixed folds for tuning and the cutoff

Test: 50,736

Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Locked

No leakage by design

Scaling lives inside the model pipelines, learned from training folds only.

Repeated answers: tested

15% of people share answers. Grouping them changed PR-AUC by 0.001 at most.

No class weighting

Same ranking, but risks inflated from 15.8% to 39%. We set the cutoff instead.

Stratified 80/20 split, seed 42 · stage 5 · 5

MODELS

Every tuned model beats the checklist; XGBoost wins every fold

Validation, mean of 5 folds	PR-AUC	ROC-AUC	Flagged for 80%
No model	0.158	0.500	100%
Paper checklist	0.325	0.756	51%
Logistic regression	0.439	0.819	39%
Random forest	0.456	0.822	39%
XGBoost (chosen)	**0.461**	**0.826**	**38%**

Rule written before the run

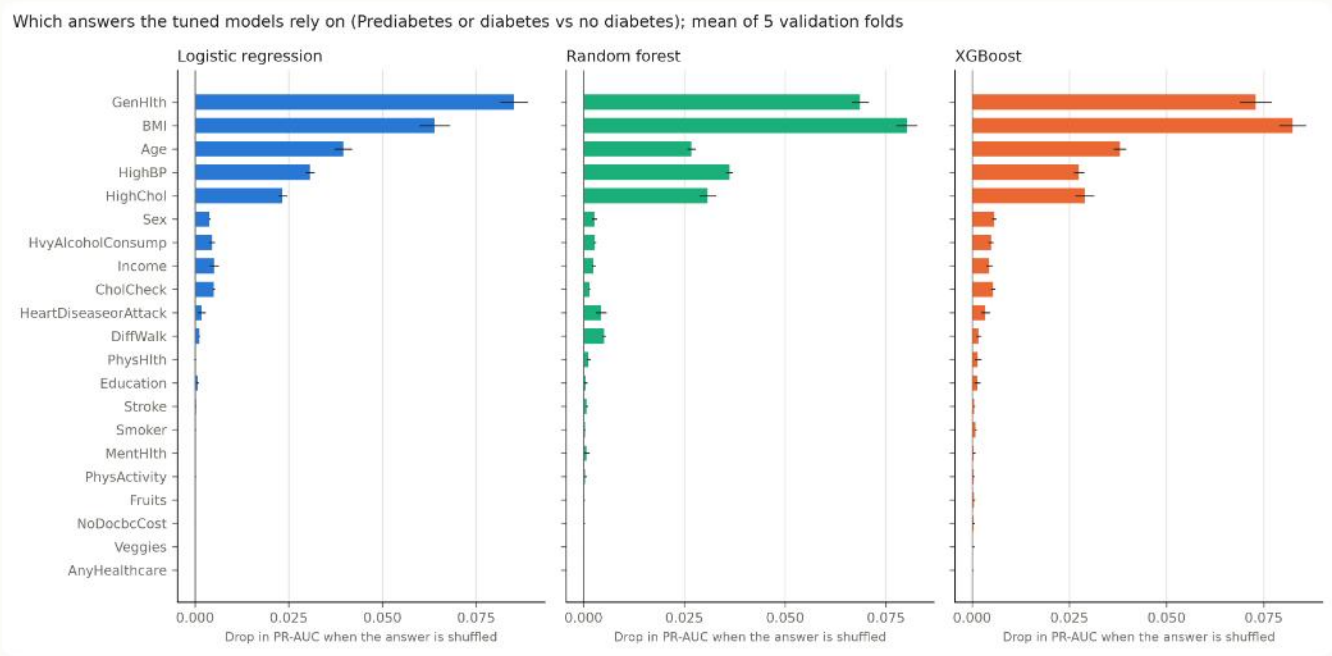
Best PR-AUC wins, unless logistic regression is not significantly worse (corrected t-test).

XGBoost beat logistic regression ($p < 0.001$) and the forest ($p = 0.005$) in all 5 folds.

Randomized search on PR-AUC, 12 / 8 / 16 candidates · stage 6 · 6

WHAT DRIVES THE PREDICTIONS

The same five answers drive every model



Adjusted odds ratios flip some stories

Physical-health days: 1.045 → 0.993, reversed

Smoking: 1.40 → 0.98, gone

Physically active: 0.50 → 0.95

Once age, BMI and general health are held fixed.

Permutation importance on validation folds; adjusted odds ratios from one statsmodels logistic regression · 7

THE SCREENING CUTOFF

We locked the cutoff before opening the test set

0.153 predicted risk; anyone at or above it is flagged

1. Out-of-fold risks on training rows only (reproduce stage 6: 0.4609 vs 0.4610)
2. Highest cutoff that still flags 80% of training cases
3. Calibration rule fixed in advance: slope 0.992, so no recalibration
4. Saved to threshold.json; rerunning gives an identical file

What other targets cost

Catch	Flagged per 1,000
70% of cases	294
80% (locked)	**380**
90% of cases	510

Out-of-fold predictions, 202,944 training rows · stage 7 · 8

FINAL RESULT · 50,736 PEOPLE THE MODEL NEVER SAW

Most cases found, far fewer false alarms than the checklist

79.4%

of cases found (95% CI 78.5–80.3%)

37.6%

of people flagged; 1 in 3 of them has the condition

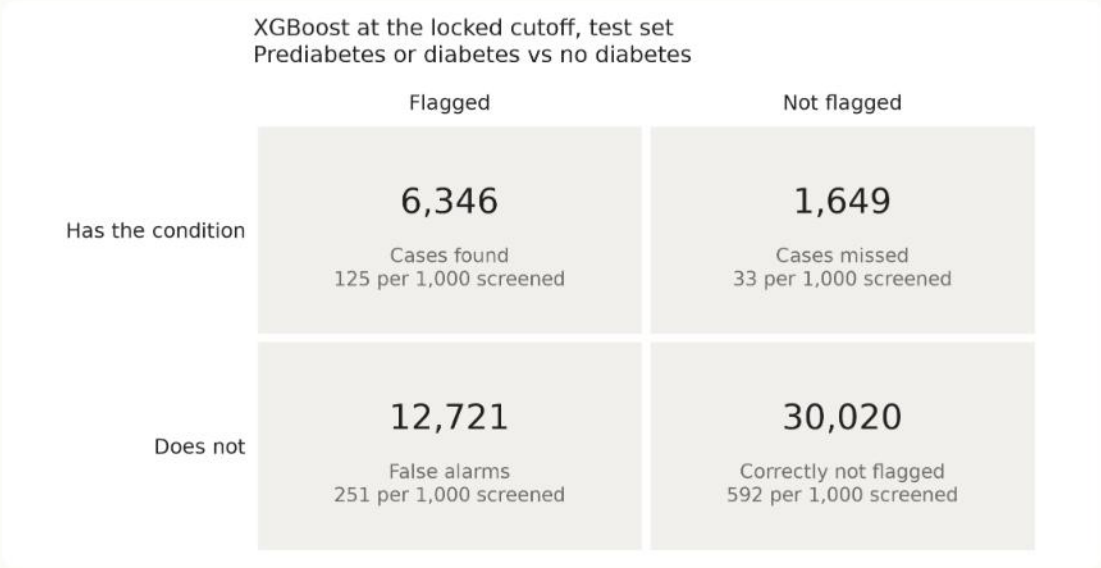
−96

false alarms per 1,000 vs the paper checklist

ROC-AUC 0.826 and PR-AUC 0.465, the same as validation · risks calibrated (slope 0.99)

WHAT IT MEANS FOR A CLINIC

Per 1,000 screened: 125 cases found, 33 missed

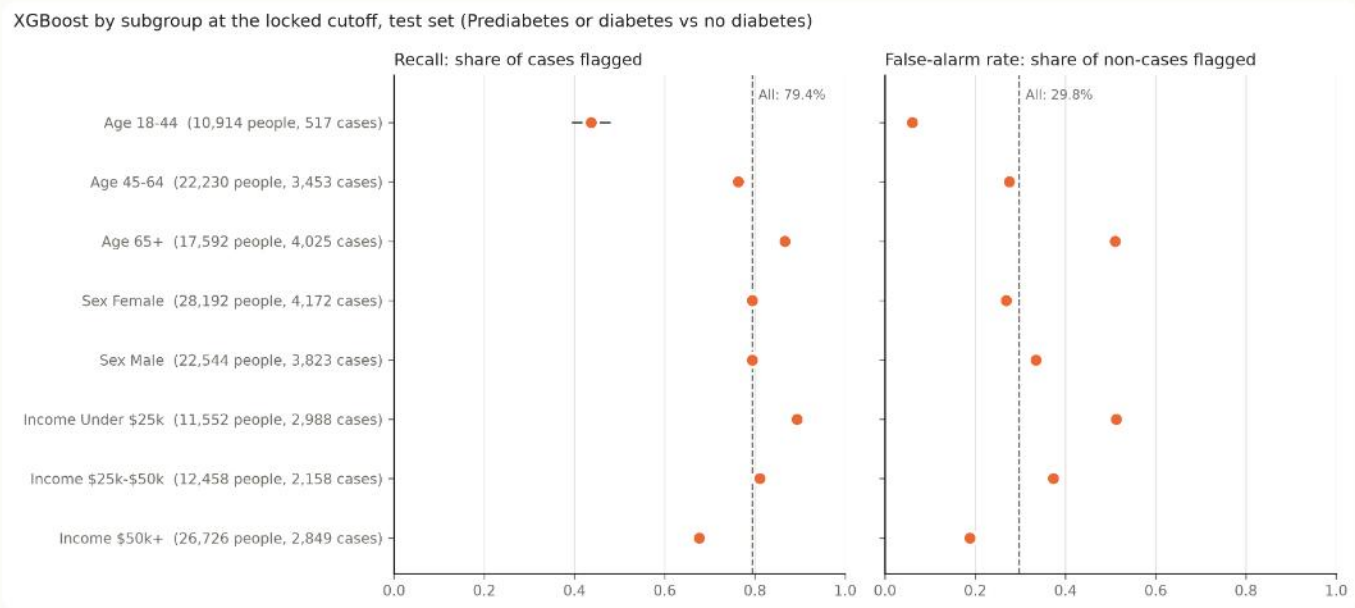


Per 1,000	Flagged	False alarms
Checklist	473	347
Logistic reg.	386	261
XGBoost	**376**	**251**

All three find about 125 of the 158 cases. A stated 30% risk means about 30 in 100 such people have the condition.

WHO GETS MISSED

It misses younger and better-off people, and prediabetes



Diabetes: 81% found

Prediabetes: 65%

Missed cases look like non-cases on paper: BMI 27.5 vs 27.7, fair or poor health 9% vs 13%.

Locked test set; Wilson 95% intervals; original survey classes from the three-class file · 11

SENSITIVITY ANALYSIS: PREDIABETES ONLY

For prediabetes alone, the signal is weak

81%

of prediabetes found (checklist: 70%)

44%

of everyone flagged to get there

4%

of flagged people report prediabetes

218,334 people without diagnosed diabetes, 2.1% with prediabetes; PR-AUC 0.062 against 0.021 for no model. Many “false alarms” may be undiagnosed prediabetes, which this survey cannot show.

Same pipeline and rules, own split and cutoff (0.0172); test set of 43,667 · 12

LIMITS

A research prototype, not a diagnostic tool

Self-reported, one moment

Identifies current, known status; predicts nothing about the future.

88% already diagnosed

The model partly learns the footprint of a diagnosis.

Doctor-contact answers

Without them: −0.03 PR-AUC, still well above the checklist.

Unequal trade-offs

One cutoff: 44% sensitivity under 45, 51% false alarms over 65.

Unweighted, one year

No national estimates; untested on other years or lab results.

Associations only

Odds ratios, adjusted or not, are not causes.

TAKEAWAY

Survey answers can find 8 in 10 cases with a fifth fewer people sent for testing than today's paper checklist

Reproducible

One command per stage; the prepared data, split, cutoff and test tables rebuild identically. 63 automated tests.

In the repository

Code, seven stage notebooks and every report: github.com/jadonsinghsumit/early-diabetes-risk-screening

Thank you. Questions?